

Yihang Zou

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Research interests: embodied AI, multimodal learning, and reliable AI systems.

EDUCATION

Duke Kunshan University & Duke University

Dual-degree undergraduate program · 2023–2027 · Expected June 2027

B.S. in Applied Mathematics and Computational Science, Computer Science track (Duke Kunshan University).

B.S. in Interdisciplinary Studies, Applied Mathematics and Computational Science (Duke University).

PUBLICATION

Y. Li, Y. Cheng, M. Ma, **Y. Zou**, N. Yang, W. Cheng, H. Li, Y. Chen, and T. Chen. “Skilled AI Agents for Embedded and IoT Systems Development.” ACM Conference on AI and Agentic Systems (CAIS 2026), pp. 1248–1252. [DOI](#).

RESEARCH EXPERIENCE

GainSight — Assistive Indoor Navigation

Undergraduate Researcher · June 2024–present · Advisor: Prof. Ming-Chun Huang

Develop a robotics platform that uses spatial audio to support nonvisual indoor navigation. Contribute to ROS 2 system integration, localization, sensor fusion, and user-centered evaluation.

IoT-SkillsBench — Embedded AI Agents

Research Assistant · September 2025–June 2026 · Advisor: Prof. Tingjun Chen

Co-developed a hardware-in-the-loop benchmark and reusable expert hardware skills for agents working with Arduino, ESP-IDF, and Zephyr. Built workflows for firmware generation, compilation, flashing, and validation on physical devices; contributed to the peer-reviewed CAIS 2026 publication.

Full-Body Motion Analysis

Undergraduate Researcher · May 2025–present · Advisor: Prof. Ming-Chun Huang

Build a monocular 2D-to-3D pose-estimation pipeline for badminton movement analysis. Combine motion-capture validation, kinematic features, and language-model feedback for structured coaching recommendations.

PROFESSIONAL EXPERIENCE

Robotics Machine Learning Intern

Chengxin Zhilian (Wuhan) Technology Co., Ltd. · July 2026–present

Work on perception and autonomy for quadruped industrial inspection, LiDAR–IMU SLAM, vision-language inference, and model deployment.

Algorithm Intern — Model Deployment

Nanjing Zhongke Nishang Technology Co., Ltd. · July–August 2025

Deployed and benchmarked large-model inference across NVIDIA, Cambricon, and Huawei accelerators. Investigated inference performance, hardware compatibility, and model adaptation.

SELECTED TECHNICAL PROJECTS

IntelliCar — Hybrid Voice-Controlled Rover

Duke ECE 655 course project · December 2025

Developed speech-to-action robot control using Vosk, Whisper, and LLM parsing, bounded JSON actions, and gyroscope-based closed-loop turning. Team project with Yuchu Guo and Yifan Xu.

Whisper Knowledge Distillation

Duke ECE 661 course project · December 2025

Built a PyTorch and Hugging Face pipeline for speech-model distillation. Compared conventional and decoupled knowledge distillation using transcription quality and model-efficiency measures.

IncogAgent — Local Privacy Middleware

Hackathon project · November 2025

Built a browser prototype that masks sensitive entities before cloud submission and restores them locally in responses, using named-entity recognition, rules, and local mappings.

TEACHING, SERVICE & AWARDS

COMPSCI 201 Tutor / Teaching Assistant · January–March 2024. Supported programming, data structures, debugging, and review sessions.

DKU Magic Club · Founder and President, August 2024–present.

DKU Varsity Badminton Team · Intercollegiate competitor, August 2023–present.

DKU Sport Complex · Associate and senior duty-group leader, August 2024–present.

2024 Gold Idea Award for assistive navigation; Second Prize, APMCM 2025; Provincial Third Prize, China Undergraduate Mathematical Contest in Modeling 2025.

TECHNICAL SKILLS

Systems: Python, C++, Java, MATLAB, Linux, Git, CUDA, ROS 2.

AI: PyTorch, Hugging Face, vision-language models, Whisper, LoRA, knowledge distillation, vLLM.

Robotics & agents: SLAM, LiDAR–IMU fusion, spatial audio, Arduino, ESP-IDF, Zephyr, Raspberry Pi, LangGraph, CAMEL, Dify.

Languages: Mandarin Chinese (native), English (fluent).